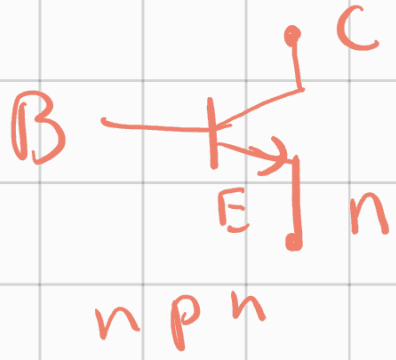
npn Junction

$$i_C = I_S e^{V_{BE}/V_T}$$

↓ saturation current

$$I_S = \frac{A_E q D_n n_{po}}{W}$$

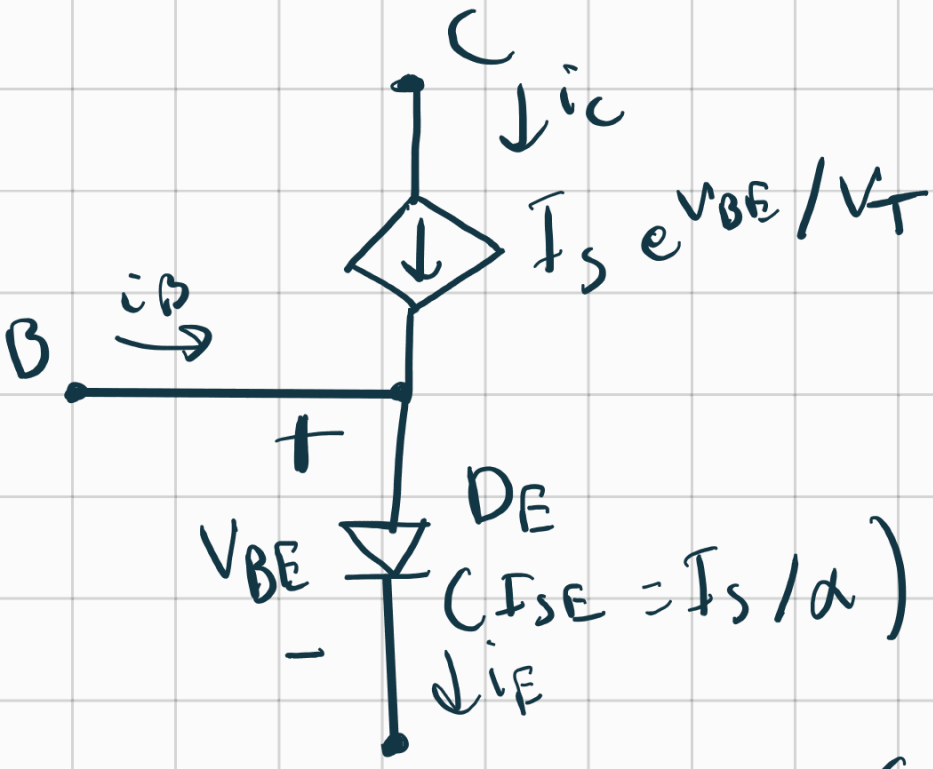


$$i_B = \frac{i_C}{\beta}$$

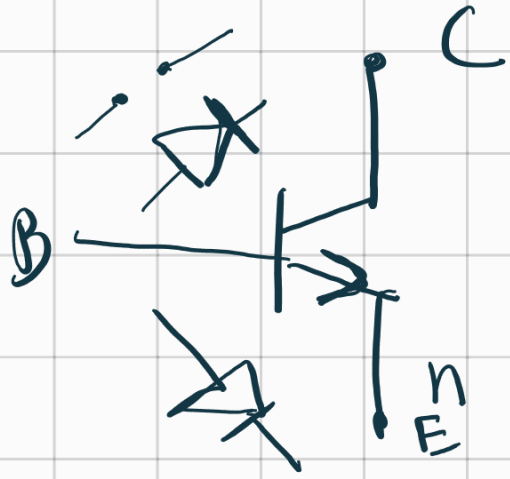
$\beta \rightarrow$ (influenced by width, and doping)

$$i_E = \frac{\beta+1}{\beta} i_C = \frac{1}{\alpha} i_C = (\beta+1) i_B$$

$$i_C = \alpha i_E$$



In npn



Active: $V_{BE} \geq 0,7$, $V_{BC} \leq 0,4V$

Cut off: $V_{BE} < 0,7$, $V_{BC} \leq 0,4V$

Saturation: $V_{BE} \geq 0,7$, $V_{BC} > 0,4$

for pnp



Active: $V_{EB} \geq 0,7V$, $V_{CB} \leq 0,4V$

Cut off: $V_{EB} < 0,7V$, $V_{CB} \leq 0,4V$

Saturation: $V_{EB} < 0,7V$, $V_{CB} > 0,4V$ //

$2 < 4$ $\quad V_C - V_B \leq 0{,}4\,V$

$-2 > -4$ $\quad V_B - V_C \geq -0{,}4$

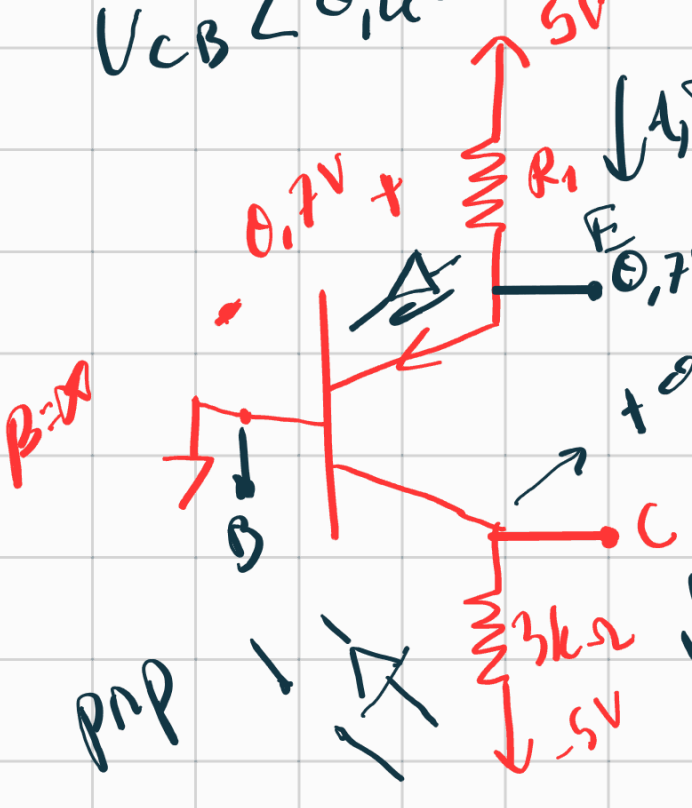
Operation in Saturation Mode

$$i_C = I_S e^{V_{BE}/V_T} - \underbrace{I_{SC} e^{V_{BC}/V_T}}_{V_{BC} > 0{,}4 \text{ için önem taşır}}$$

Base current in sat region: $i_B = \frac{I_S}{\beta} e^{V_{BE}/V_T} + I_{SC} e^{V_{BC}/V_T}$

forced $\beta = \left.\frac{i_C}{i_B}\right|_{sat} \leq \beta$

$V_{CB} < 0{,}4$ olmalı



Determine the value of R_1 such that Q_1 operate in active mode

$$i_E = \frac{5 - 0{,}7}{R_1} = \frac{4{,}7}{R_1}$$

$$i_E = \frac{\beta}{\beta+1} i_C = i_C \quad (\beta = \infty)$$

$V_{CB} = 0{,}4\,V$

$$i_C = \frac{+0{,}4 + 5}{3k} = \frac{5{,}4}{3k} = 1{,}8\,mA$$

$R_1 \geq \frac{4,7}{18mA} \geq 2,3k\Omega$

Çünkü

$V_{CB} < 0,4$
eğer akım artarsa V_{CB} artar
yani direnç azalırsa

//madirenç artarsa akım da azalır ve böylece çözmüş oluruz.//